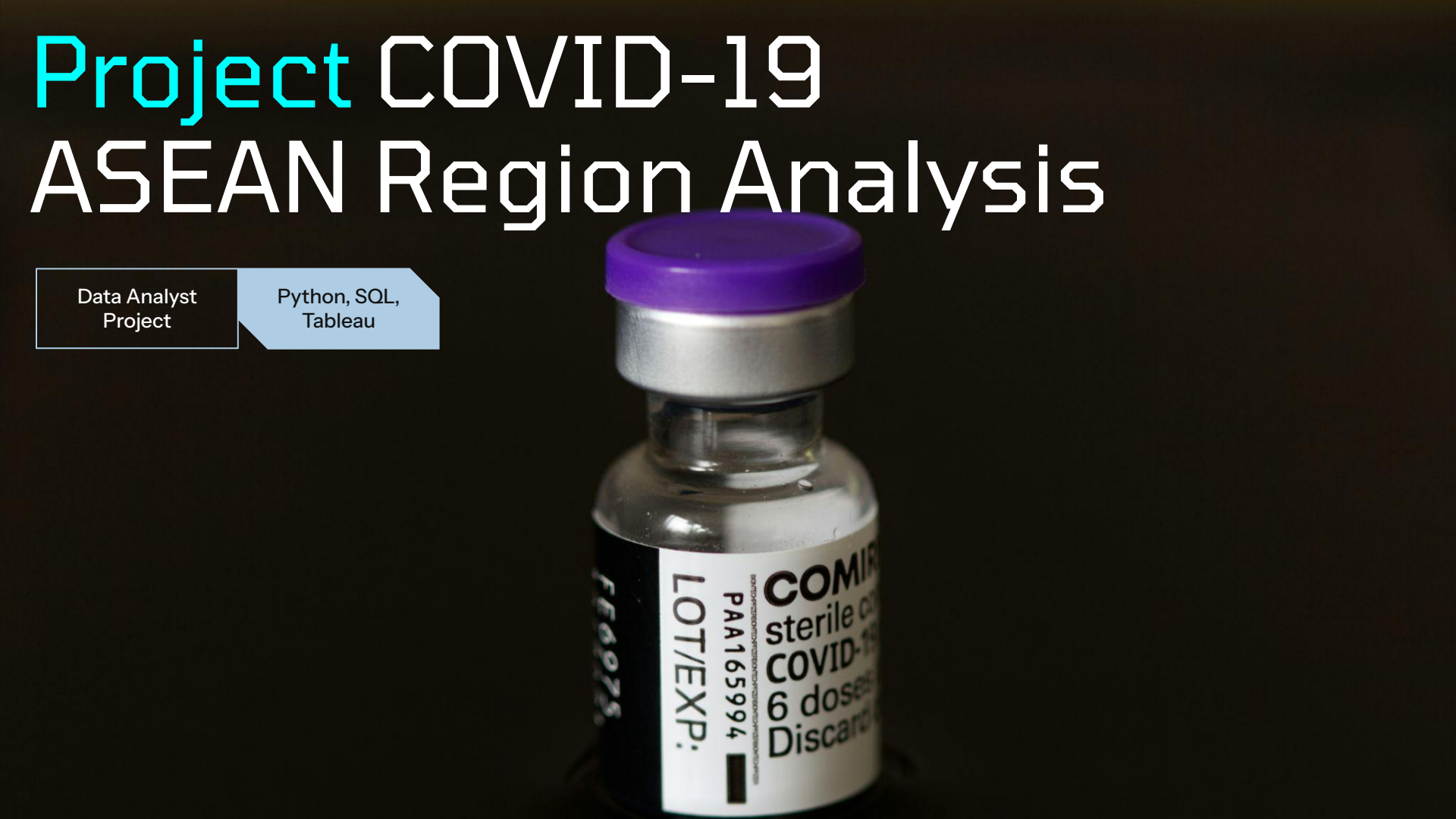
Project COVID-19 ASEAN Region Analysis

Data Analyst Project

Python, SQL, Tableau



My Identity

Dean Gery Pasamba

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Hello everybody, my name is Dean Gery Pasamba. I'm highly enthusiastic on data role and interested in starting a career as a data analyst, business analyst, or even a data scientist. In addition to continuously learning about the data field, I also have software engineering skills such as web and mobile development, along with knowledge about security engineering.

Overview Projects

Covid-19 has been one of the greatest disasters in the recent years. It caused a lot of deaths and was recognized as one of the most deadly diseases of our time. It spread widely from 2019 till 2023. In 2020, it officially became a pandemic, and that global emergency was later declared over in 2023. The solution came from vaccines that were developed over almost 2 years. But despite the rapid solution, it has taken a lot of live. This project will analyze the covid-19 specifically on ASEAN countries. Hope you like it !😊.

Datasets

OUR World In Data Covid-19 Dataset 2020-2026 : https://ourworldindata.org/covid-deaths

This Datasets contains 567121 rows of data with 61 features. To make it compatible with Mysql, I split the data into two separate files which are CovidDeaths.csv and CovidVaccinations.csv.

Our World in Data

Tech Stack

Python

Python is used to extract, transform, load and clean the dataset.

SQL

SQL is mainly used for its queries to transform the dataset to uncover insights.

Tableau

Tableau is used for visualization and gaining the best insights to communicate.

Analysis?

As with most typical data analysis projects, I defined several key questions to answer. These questions mainly focus on the impact of COVID-19 on ASEAN countries and how they handle it. I formulated it during the data exploration query process, right after the Python ETL step.

Python (Extract, Transform, Load, and Clean)

```
[ ]  df_dea = pd.read_csv('compact.csv')

[ ]  len(df_dea)

     567120
```

The first thing I did was extract and preprocess the dataset in python (colab ipynb). I looked carefully into the data, where I found there are lots of NaN values that needed to be filled so it could be processed in MySql.

```
[ ]  df_dea.head(10)
```

	country	date	total_cases	new_cases	new_cases_smoothed	total_cases_per_million	new_cases_per_million	new_cases_smoothed_per_milli
0	Afghanistan	2020-01-01	NaN	NaN	NaN	NaN	NaN	Na
1	Afghanistan	2020-01-02	NaN	NaN	NaN	NaN	NaN	Na
2	Afghanistan	2020-01-03	NaN	NaN	NaN	NaN	NaN	Na

Python (Extract, Transform, Load, and Clean)

```python
pd.set_option('display.max_seq_items', None)
pd.set_option('display.max_rows', None)
```

```python
df_dea['country'].unique()
```

```
<ArrowStringArray>
[
                                    'Afghanistan',
                                         'Africa',
                                        'Albania',
                                        'Algeria',
                                 'American Samoa',
                                        'Andorra',
                                         'Angola',
                                       'Anguilla',
                             'Antigua and Barbuda',
                                      'Argentina',
                                        'Armenia',
                                          'Aruba',
                                           'Asia',
                                 'Asia excl. China',
```

Before fixing the data, I rechecked again to make sure the country values were correct. Then, I found a lot of entries that weren't actually countries, such as 'world', and 'European Union (27)'. So I stored them in variables to clean them out first before filling the NaN values.

```python
isocode_owid = df_dea['code'].astype(str).str.startswith('OWID_')
isocode_kosong = df_dea['code'].isna()
```

```python
pd.reset_option('display.max_seq_items')
pd.reset_option('display.max_rows')
```

```python
df_non_negara = df_dea[isocode_kosong | isocode_owid]
```

```python
df_non_negara['country'].unique().tolist()
```

Python (Extract, Transform, Load, and Clean)

```python
df_non_negara['country'].unique().tolist()
```

```
['Africa',
 'Asia',
 'Asia excl. China',
 'England',
 'England and Wales',
 'Europe',
 'European Union (27)',
 'High-income countries',
 'Kosovo',
 'Low-income countries',
 'Lower-middle-income countries',
 'North America',
 'Northern Ireland',
 'Oceania',
 'Scotland',
 'South America',
 'Summer Olympics 2020',
 'Transnistria',
 'Upper-middle-income countries',
 'Wales',
 'Winter Olympics 2022',
 'World',
 'World excl. China',
 'World excl. China and South Korea',
 'World excl. China, South Korea, Japan and Singapore']
```

```python
hapus_data = ['Africa',
 'Asia',
 'Asia excl. China',
 'England',
 'England and Wales',
 'Europe',
 'European Union (27)',
 'High-income countries',
 'Kosovo',
 'Low-income countries',
 'Lower-middle-income countries',
 'North America',
 'Northern Ireland',
 'Oceania',
 'Scotland',
 'South America',
 'Summer Olympics 2020',
 'Transnistria',
 'Upper-middle-income countries',
 'Wales',
 'Winter Olympics 2022',
 'World',
 'World excl. China',
 'World excl. China and South Korea',
 'World excl. China, South Korea, Japan and Singapore']
```

```python
df_clean = df_dea[~df_dea['country'].isin(hapus_data)].copy()
```

To clean all of the false country values, I did it manually by checking and storing them into a variable. After that, I used exclusion filtering to remove false values and created a copy of dataframe named df_clean.

Python (Extract, Transform, Load, and Clean)

```python
df_clean.info()
```

```
<class 'pandas.DataFrame'>
Index: 522849 entries, 0 to 567119
Data columns (total 61 columns):
 #   Column                                     Non-Null Count   Dtype
---  ------                                     --------------   -----
 0   country                                    522849 non-null  str
 1   date                                       522849 non-null  str
 2   total_cases                                516896 non-null  float64
 3   new_cases                                  513012 non-null  float64
 4   new_cases_smoothed                         511888 non-null  float64
 5   total_cases_per_million                    516896 non-null  float64
 6   new_cases_per_million                      513012 non-null  float64
 7   new_cases_smoothed_per_million             511888 non-null  float64
 8   total_deaths                               516896 non-null  float64
```

```python
kolom_analisis = [
    'continent', 'country', 'date', 'population',
    'total_cases', 'new_cases', 'total_deaths', 'new_deaths',
    'new_vaccinations', 'stringency_index', 'reproduction_rate',
    'gdp_per_capita', 'hospital_beds_per_thousand', 'total_deaths_per_million',
    'people_fully_vaccinated_per_hundred', 'new_deaths_smoothed_per_million',
    'median_age', 'diabetes_prevalence'
]
```

```python
df_final = df_clean[kolom_analisis].copy()
```

```python
df_final = df_final.rename(columns={'country': 'location'})
```

Next, I filtered the data to keep only the useful features. After that I copied it into a new data frame called df_final, and also I renamed the ‘country’ column into ‘location’ to make further processing easier.

Python (Extract, Transform, Load, and Clean)

```python
missing_percentages = (df_final.isnull().sum() / len(df_clean)) * 100
missing_percentages = missing_percentages.sort_values(ascending=False)

print("Persentase Missing Values tiap kolom:\n")
print(missing_percentages.head(20))
```

```
Persentase Missing Values tiap kolom:

new_vaccinations                    89.999790
people_fully_vaccinated_per_hundred 88.458236
reproduction_rate                   65.050330
stringency_index                    61.429782
hospital_beds_per_thousand          36.427343
gdp_per_capita                      18.170064
diabetes_prevalence                 10.919023
new_deaths_smoothed_per_million      1.903609
new_cases                            1.881423
new_deaths                           1.688059
total_deaths_per_million             1.138570
total_cases                          1.138570
total_deaths                         1.138570
population                           0.426127
median_age                           0.426127
continent                            0.000000
location                             0.000000
date                                 0.000000
dtype: float64
```

```python
missing_percentages = (df_final.isnull().sum() / len(df_clean)) * 100
missing_percentages = missing_percentages.sort_values(ascending=False)

print("Persentase Missing Values tiap kolom:\n")
print(missing_percentages.head(20))
```

```
Persentase Missing Values tiap kolom:

continent                           0.0
location                            0.0
date                                0.0
population                          0.0
total_cases                         0.0
new_cases                           0.0
total_deaths                        0.0
new_deaths                          0.0
new_vaccinations                    0.0
stringency_index                    0.0
reproduction_rate                   0.0
gdp_per_capita                      0.0
hospital_beds_per_thousand          0.0
total_deaths_per_million            0.0
people_fully_vaccinated_per_hundred 0.0
new_deaths_smoothed_per_million     0.0
median_age                          0.0
diabetes_prevalence                 0.0
dtype: float64
```

Next up, I began to fill the NaN values. In this step, I checked the percentage of all features that had missing values. Then, I filled all of the missing values with 0.

```python
df_final = df_final.fillna(0)
```

```python
df_final.isnull().sum().sum()
```

```
np.int64(0)
```

Python (Extract, Transform, Load, and Clean)

```python
kolom_deaths = [
    'continent', 'location', 'date', 'population',
    'total_cases', 'new_cases', 'total_deaths', 'new_deaths',
    'reproduction_rate', 'stringency_index'
]
df_CovidDeaths = df_final[kolom_deaths].copy()
```

```python
kolom_vaccinations = [
    'continent', 'location', 'date', 'population',
    'new_vaccinations', 'people_fully_vaccinated_per_hundred',
    'gdp_per_capita', 'hospital_beds_per_thousand',
    'median_age', 'diabetes_prevalence'
]
df_CovidVaccinations = df_final[kolom_vaccinations].copy()
```

```python
df_CovidDeaths['date'] = pd.to_datetime(df_CovidDeaths['date'])
```

```python
df_CovidVaccinations['date'] = pd.to_datetime(df_CovidVaccinations['date'])
```

Finally, I split the data into two separate files CovidDeaths and CovidVaccinations. For each file, I only selected the features that were relevant to it. To make sure the data was optimized, I also convert date column into proper datetime format.

```python
df_CovidDeaths.to_csv('CovidDeaths.csv', index=False)
df_CovidVaccinations.to_csv('CovidVaccinations.csv', index=False)
```

Project Questions

01 How do total cases and death rates of COVID-19 in ASEAN compared to the world ?

SQL Exploration (Global Number)

```sql
SELECT
    SUM(new_cases) AS total_cases,
    SUM(CAST(new_deaths AS SIGNED)) AS total_deaths,
    SUM(CAST(new_deaths AS DECIMAL)) / NULLIF(SUM(new_cases), 0) * 100 AS DeathPercentage
FROM Portfolio_Project.CovidDeaths
WHERE continent IS NOT NULL AND continent != ''
ORDER BY 1, 2;
```

To answer this question, we needed information about global COVID-19 total cases and total deaths. To do that, I imported the dataset into MySQL and used SQL query to extract the data. Here, I summed up the new_cases column to calculate the total cases, and summed new_deaths to get the total deaths. I also calculated the death percentage using a standard formula, and then ordered the results.

SQL Exploration (ASEAN Region Number)

```sql
SELECT
    'ASEAN' AS Region,
    SUM(new_cases) AS Total_Cases,
    SUM(new_deaths) AS Total_Deaths,
    SUM(new_deaths) / NULLIF(SUM(new_cases), 0) AS Death_Percentage
FROM Portfolio_Project.CovidDeaths
WHERE location IN (
    'Brunei', 'Cambodia', 'Indonesia', 'Laos', 'Malaysia',
    'Myanmar', 'Philippines', 'Singapore', 'Thailand', 'Vietnam'
);
```

To compare the global numbers with ASEAN countries, I applied the same logic but declared the value 'ASEAN' as a new region column. I also filtered the location to include only ASEAN countries. After that, I saved the output table as a CSV file to be visualized in Tableau.

Tableau (Global & ASEAN Number)

Global Number (2020-2026)

Total Cases	Total Deaths	Death Percentage
778,782,618	7,113,812	0.91%

ASEAN Number (2020 - 2026)

Region	Total Cases	Total Deaths	Death Percentage
ASEAN	37,768,546	369,969	0.98%

The first results show that globally, there were nearly 779 million COVID-19 cases and 7.1 million deaths, resulting in a global death rate of about 0.91%. When we look at the ASEAN region, despite having nearly 38 million total cases, the death rate was 0.98%. The number is slightly higher, which indicates that ASEAN might have faced specific challenges that made the virus slightly more deadly here.

02 Which ASEAN countries had the highest infection rate from year to year ?

SQL Exploration (Population Infected Percentage)

```sql
SELECT
    Location,
    Population,
    MAX(total_cases) AS HighestInfectionCount,
    MAX((total_cases * 1.0 / population)) * 100 AS PercentPopulationInfected
FROM Portfolio_Project.CovidDeaths
GROUP BY Location, Population
ORDER BY PercentPopulationInfected DESC;
```

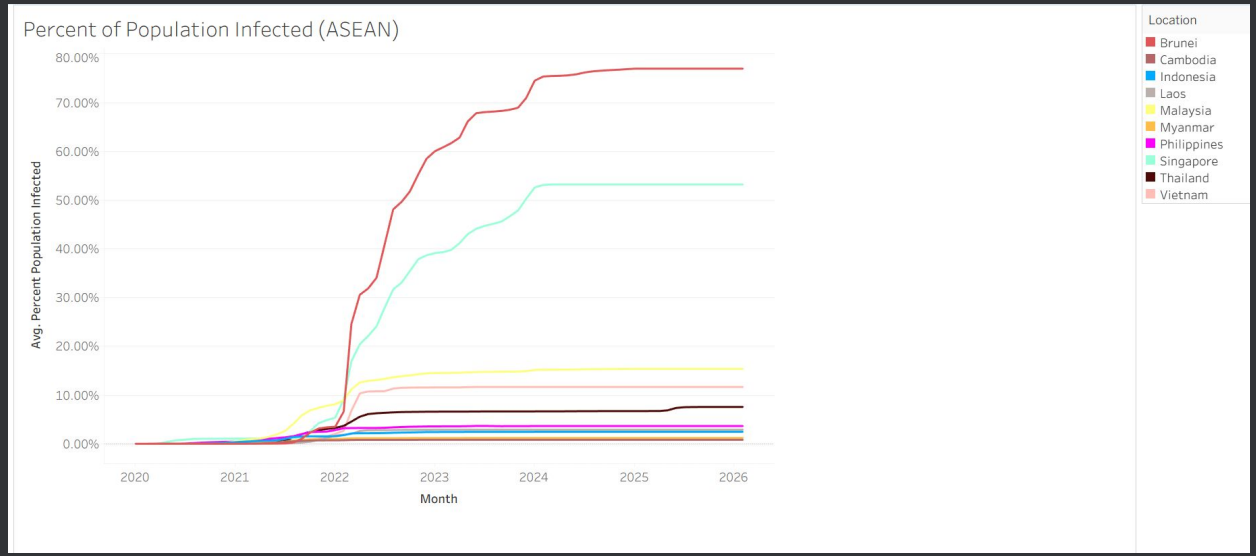
To answer this question, I used a query to show the percentage of the population infected. In this query, I selected the location, population and total_cases as HighestInfection Count to extract the necessary data.

SQL Exploration (Pop Infected Percentage Time Series)

```sql
SELECT
    Location,
    Population,
    date,
    MAX(total_cases) AS HighestInfectionCount,
    MAX((total_cases * 1.0 / population)) * 100 AS PercentPopulationInfected
FROM Portfolio_Project.CovidDeaths
GROUP BY Location, Population, date
ORDER BY PercentPopulationInfected DESC;
```

Additionally, I wanted to create a geographic map visualization. Using the same logic, I added a date column to generate time-series data, which serves as the foundation for the map

Tableau (Geo Map & Percent Pop Infected)



From the line chart we can see that country with the highest infection rate relative to its population is Brunei. This is highly likely because Brunei has a relatively small population. On the other hand, Cambodia shows the lowest infection rate on the graph.

Tableau (Geo Map & Percent Pop Infected)

The geographic map provides a clearer spatial view of these insights. We can easily compare the numbers, highlighting Brunei with 76.98% of its population infected, contrasting sharply with Cambodia at just 0.81%.

Project Questions

03

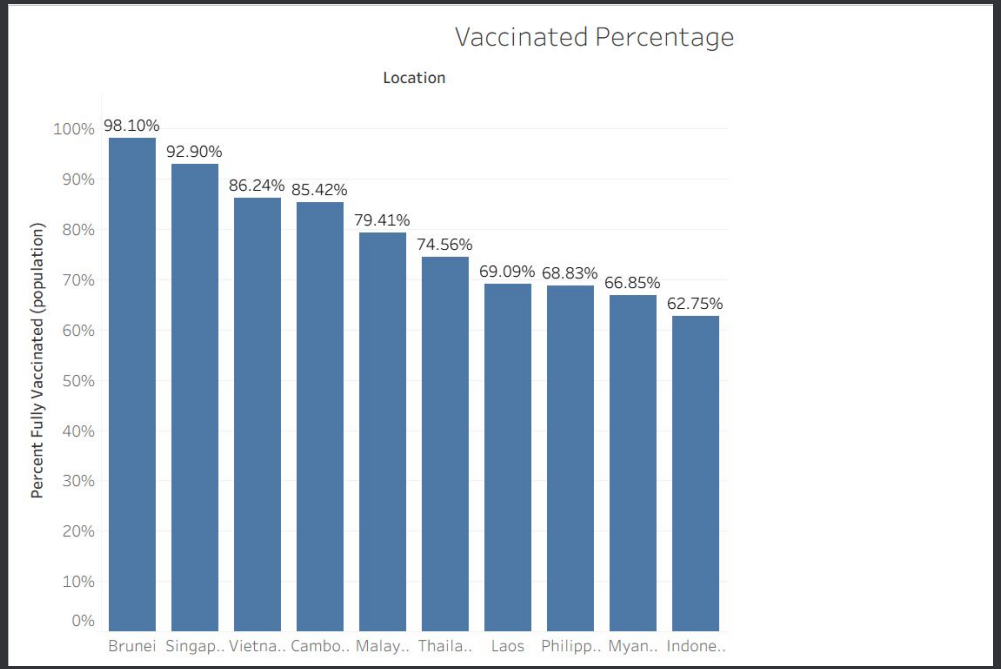
How is vaccination distribution in ASEAN countries, which countries had the highest vaccination and which one is the lowest ?

SQL Exploration (ASEAN Vaccinated Number)

```sql
SELECT * from covidvaccinations;
SELECT
    dea.location,
    MAX(dea.population) AS Total_Population,

    -- Mengambil persentase dari tabel vaksinasi (vac)
    MAX(vac.people_fully_vaccinated_per_hundred) / 100 AS Percent_Fully_Vaccinated,

    -- Menghitung jumlah orang dari gabungan tabel vac dan dea
    (MAX(vac.people_fully_vaccinated_per_hundred) / 100) * MAX(dea.population) AS Total_People_Vaccinated

FROM Portfolio_Project.CovidDeaths dea
JOIN Portfolio_Project.CovidVaccinations vac
    ON dea.location = vac.location
    AND dea.date = vac.date
WHERE dea.location IN (
    'Brunei', 'Cambodia', 'Indonesia', 'Laos', 'Malaysia',
    'Myanmar', 'Philippines', 'Singapore', 'Thailand', 'Vietnam'
)
GROUP BY dea.location
ORDER BY Percent_Fully_Vaccinated DESC;
```

For this part, I joined the two datasets to find deeper insights. By taking the population from the deaths table and the vaccination rate per hundred from the vaccines table, I could calculate the total population, total vaccinated people, and the vaccination percentage. I joined them by both location and date, filtering the data to only show ASEAN nations.

Tableau (ASEAN Vaccinated Pop Percentage)



I created the visualization based on vaccination to population percentage and location. As we can see, Brunei leads the region at over 98%, whereas Indonesia surprisingly has the lowest rate relative to its population at roughly 62%. If we calculate the regional average, it sits at about 78%, which tells us that overall vaccination in ASEAN is pretty good, but the distribution varies significantly.

04

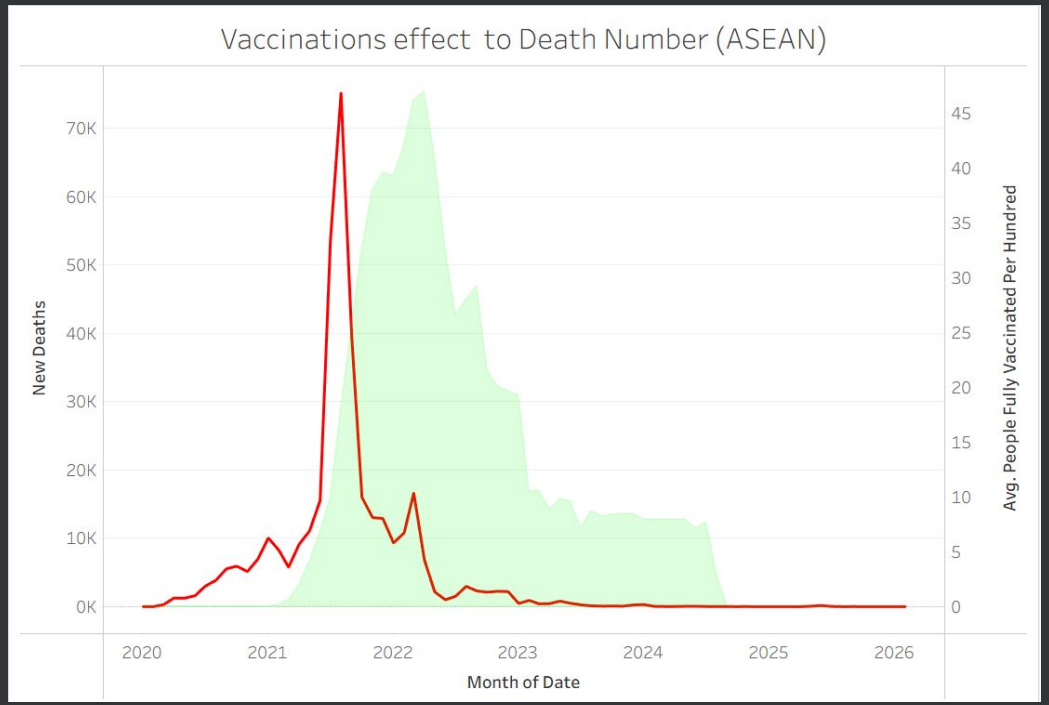
Did the rollout of vaccinations effectively reduce the number of new deaths?

SQL Exploration (Vaccination & Deaths)

```sql
SELECT
    dea.location,
    dea.date,
    vac.people_fully_vaccinated_per_hundred,
    dea.new_deaths
FROM Portfolio_Project.CovidDeaths dea
JOIN Portfolio_Project.CovidVaccinations vac
    ON dea.location = vac.location
    AND dea.date = vac.date
WHERE dea.continent IS NOT NULL AND dea.continent != ''
ORDER BY dea.location, dea.date;
```

To answer the question and analyze it, I pulled the location, date, vaccination, and daily death metrics. I then joined the two tables by matching their locations and dates to get a clear time-series view.

Tableau (Vaccination to Death Number)



By using tableau, I combined a line and area chart to show how vaccines impacted the death rate. As we can clearly see, once the vaccination numbers started rising, the death toll went down sharply and remained low. This gives us clear visual evidence that the vaccination rollout was extremely effective in saving lives.

05

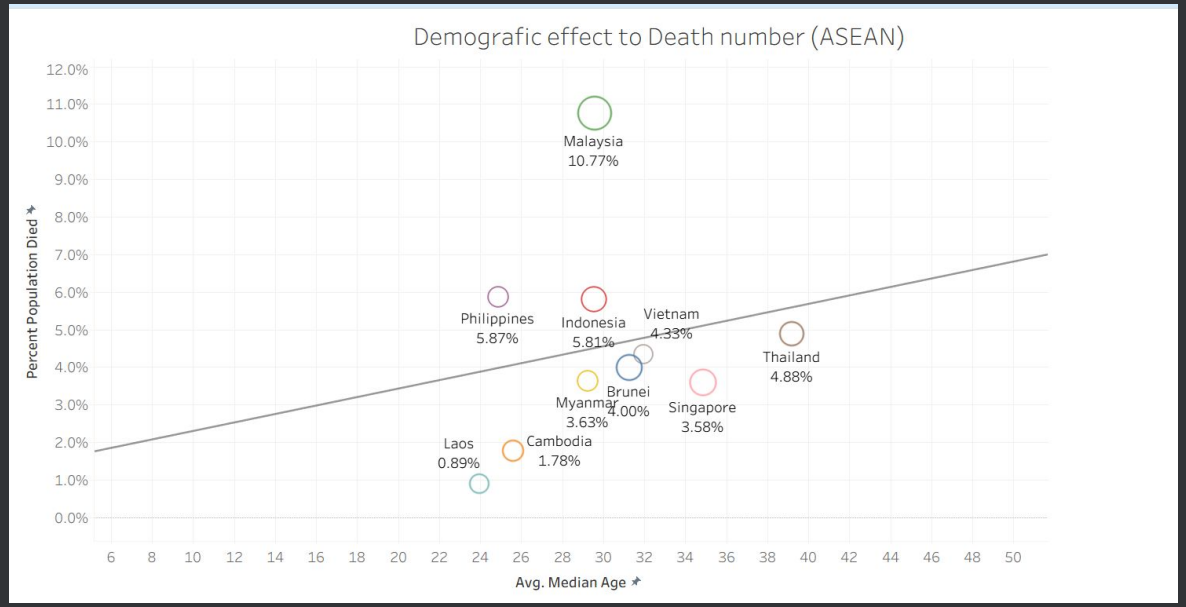
Is there any correlation between a country's median age and its population death rate?

SQL Exploration (Demografic Effect)

```sql
SELECT dea.location,
    MAX(vac.median_age) AS median_age,
    MAX(vac.diabetes_prevalence) AS diabetes_prevalence,
    (MAX(dea.total_deaths)/ MAX(dea.population)) * 100 AS PercentPopulationDied
FROM Portfolio_Project.CovidDeaths dea
JOIN Portfolio_Project.CovidVaccinations vac
    ON dea.location = vac.location
    AND dea.date = vac.date
WHERE dea.continent IS NOT NULL AND dea.continent != ''
GROUP BY dea.location
ORDER BY PercentPopulationDied DESC;
```

To explore the answer, I selected the median age and diabetes prevalence columns, and calculated the percentage of the population that died (total deaths divided by population). I joined the datasets on location and date to prepare these results for Tableau.

Tableau (Demografic Effect)



In tableau, I manually filtered ASEAN countries and used scatter plot with linear regression to to examine the correlation. From the results we can see an upward line indicates positive correlation. But it's clearly not the whole factors. We can see a major outlier where the highest death rate (Malaysia) occurred in a population with a median age of around 30. So, while age is a significant factor, it's definitely not the only one at play.

06

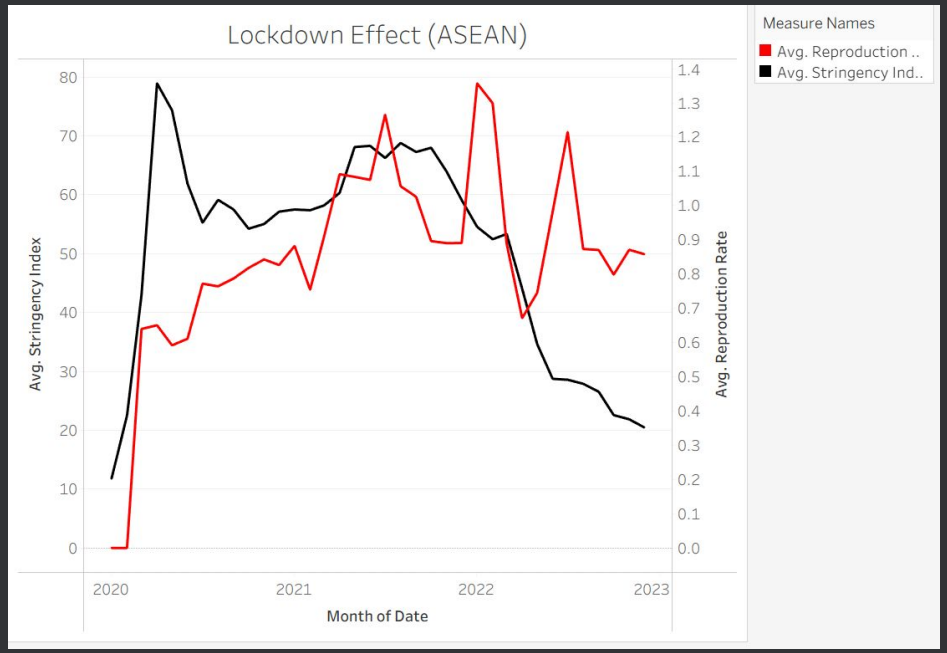
Did government lockdown policies effectively lower the virus's reproduction rate?

SQL Exploration (Lockdown & Infection Rate)

```sql
SELECT
    location,
    date,
    stringency_index,
    reproduction_rate,
    new_cases
FROM Portfolio_Project.CovidDeaths
WHERE continent IS NOT NULL AND continent != ''
AND stringency_index > 0
ORDER BY location, date;
```

For this section, I extracted the location, date, stringency_index (lockdown index), reproduction_rate, and new_cases. This data will be exported to Tableau, where I will use visualizations to compare these metrics and analyze the actual effect of lockdown policies on the spread of the virus.

Tableau (Lockdown Effect)



The chart compares the stringency index (black) with the reproduction rate (red). High stringency in 2020-2021 successfully prevented major spikes in reproduction. As lockdowns eased by 2023, fluctuations continued. This indicates lockdowns were effective in controlling, though not completely halting, the spread of the virus.

07

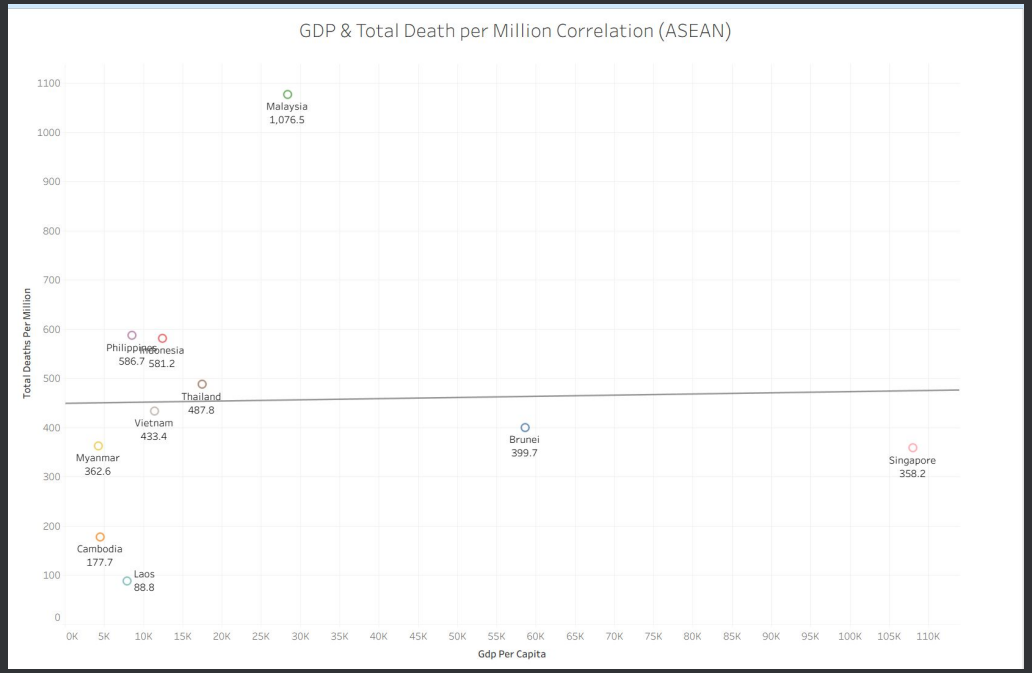
Does a country's economic status (GDP per capita) correlate with its total deaths per million?

SQL Exploration (GDP and Total Death)

```sql
SELECT dea.location,
    MAX(vac.gdp_per_capita) AS gdp_per_capita,
    MAX(vac.hospital_beds_per_thousand) AS hospital_beds_per_thousand,
    (MAX(dea.total_deaths)/ MAX(dea.population)) * 1000000 AS total_deaths_per_million
FROM Portfolio_Project.CovidDeaths dea
JOIN portfolio_project.CovidVaccinations vac
    ON dea.location = vac.location
    AND dea.date = vac.date
WHERE dea.continent IS NOT NULL AND dea.continent != ''
GROUP BY dea.location
ORDER BY gdp_per_capita DESC;
```

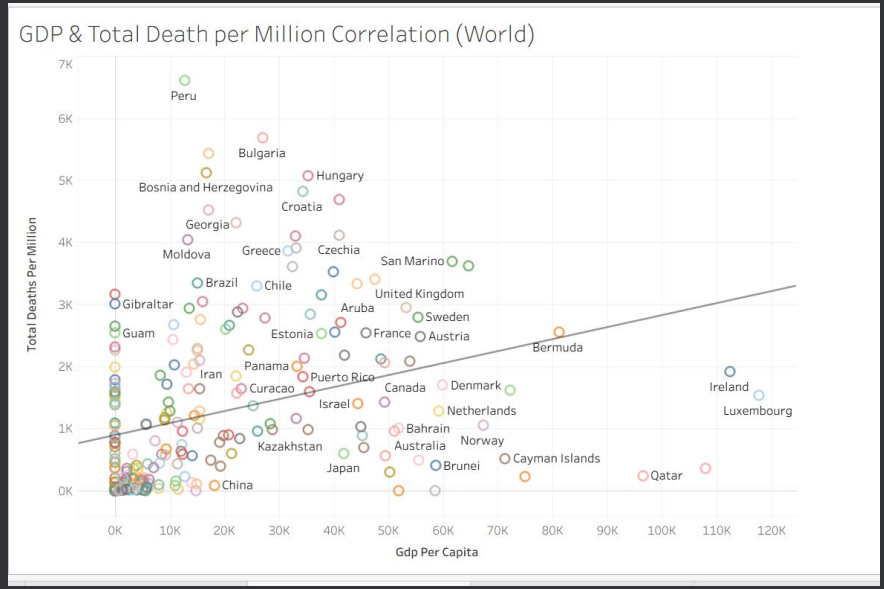
To answer this question, I extracted the GDP per capita and hospital beds per thousand metrics. Additionally, I calculated the total deaths per million by dividing the maximum total deaths by the population. The resulting dataset will be exported and visualized in Tableau.

Tableau (ASEAN GDP & Total Death Correlation)



To find the correlation I applied linear regression. Here we can see that in ASEAN, GDP isn't strongly correlated with the death rate. This is likely due to the small sample size and the fact that few countries in the region have a GDP above $50k. However, we can observe that the wealthiest nations, Singapore and Brunei, successfully maintained relatively low mortality rates, averaging around 378.95 deaths per million. To get a clearer picture of how GDP impacts, we would need to look at a larger, global dataset.

Tableau (World GDP & Total Death Correlation)



In this here, I applied the same visualization using global data. Surprisingly, the scatter plot shows a positive correlation between GDP per capita and total deaths per million. This means that, on average, countries with higher GDPs actually reported higher mortality rates. While this might seem counterintuitive, it is likely because wealthier nations have more robust testing and data reporting infrastructures, meaning they captured and recorded more cases accurately. Additionally, higher-GDP countries tend to have older populations, which, as we saw earlier, are more vulnerable to the virus.

Project highlights

Question Answer

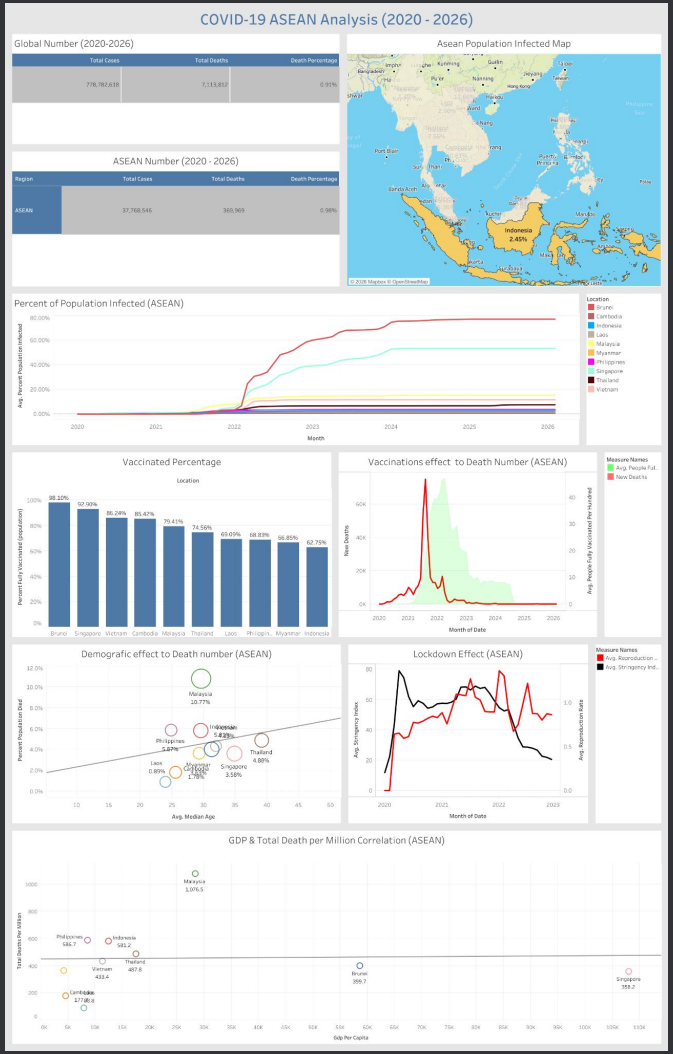
01	How do total cases and death rates of COVID-19 in ASEAN compared to the world ?	Globally, the death rate stood at 0.91%. However, the ASEAN region recorded a slightly higher death rate of 0.98%. This indicates that ASEAN might have faced specific regional challenges that made the virus slightly more deadly compared to the global average
02	Which ASEAN countries had the highest infection rate from year to year ?	Brunei recorded the highest infection rate, with 76.98% of its population infected, which is likely due to its relatively small population size. In stark contrast, Cambodia showed the lowest infection rate at just 0.81%.
03	How is vaccination distribution in ASEAN countries, which countries had the highest vaccination and which one is the lowest ?	The overall vaccination distribution in ASEAN is pretty good, averaging around 78%. Brunei leads the region with a fully vaccinated rate of over 98%, while Indonesia surprisingly has the lowest rate relative to its massive population at roughly 62.75%.

Question Answer

04	Did the rollout of vaccinations effectively reduce the number of new deaths?	Yes, absolutely. The data clearly shows that once vaccination numbers started rising, the death toll went down and remained low, providing strong evidence that the vaccination was extremely effective.
05	Is there any correlation between a country's median age and its population death rate?	Yes, there is a positive correlation, meaning older populations generally face higher death rates. However, age is not the only factor at play; for instance, Malaysia experienced the highest death rate despite having a median age of around 30.
06	Did government lockdown policies effectively lower the virus's reproduction rate?	High stringency measures successfully prevented major spikes in the reproduction rate during 2020-2021. While lockdowns did not completely halt the virus, they were effective in controlling its spread.
07	Does a country's economic status (GDP per capita) correlate with its total deaths per million?	In ASEAN, the correlation is weak due to a small sample size and a lack of high-GDP countries. Globally, however, there is a positive correlation, wealthier countries reported higher death rates, highly likely due to having more robust testing and accurate data reporting infrastructures.

Dashboard Visualization

https://public.tableau.com/app/profile/gerynsb.deangerypasamba/viz/BookCovid19ASEAN/Dashboard1



Conclusion

The COVID-19 pandemic impacted the **ASEAN region** with unique challenges, resulting in a slightly higher **mortality rate (0.98%)** compared to the global average (0.91%). While early **lockdown policies** were essential in **controlling** the virus's **reproduction rate**, the mass rollout of **vaccines** proved to be the **most** decisive **factor** in **driving down** the **death toll** across the region. Furthermore, this analysis highlights that beyond **demographics** like **age**, a **country's economic status (GDP)** heavily **influenced data accuracy**, **wealthier nations** simply had **better infrastructure** to test and **report cases**. Ultimately, **surviving** the **pandemic** required a **combination** of **strict** early **interventions** and **aggressive vaccination** strategies.

Source Code

https://github.com/gerynsb/Covid-19_Data_Analysis_Projects

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Thank you

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